2022A F=ma Exam: Problem 1

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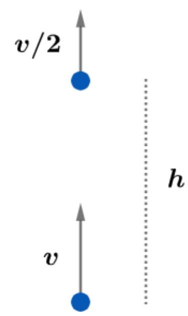

Setting the gravitational potential energy reference at the starting location, the initial energy of the projectile is

$$E_i = \frac{1}{2}mv^2$$

The energy of the projectile after it goes up by height h is

$$E_h = \frac{1}{2}m\left(\frac{v}{2}\right)^2 + mgh$$

Finally, the energy at the highest point is

$$E_f = mgH$$

By conservation of energy, $E_i = E_h$,

$$\frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{v^2}{4}\right) + mgh$$

$$\frac{1}{2}m\left(\frac{3v^2}{4}\right) = \frac{3}{4}E_i = mgh$$

Setting $E_i = 4mgh/3$ equal to $E_f = mgH$ yields

$$H = \frac{4h}{3}$$

so the answer is $\boxed{\text{B}}$.